

# CS 3015 – Robotics Technology

Fall 2024

## Assignment II – Requirement Document

Faculty: Umarah Qaseem

Assigned on: 25th October 2024

### Instructions:

1. **Guidelines Compliance:** Any violation of the provided guidelines will result in a score of zero for the assignment.
2. **Handwritten Submission:** The assignment must be completed in your handwriting. Printed copy can only be used where explicitly mentioned.
3. **No Email or Soft copy Submissions:** Under no circumstances will submissions via email or digital format be accepted. All assignments must be submitted in handwritten format only.
4. **Individual Work:** This is an individual assignment. Collaboration or copying will not be tolerated.
5. **Plagiarism:** STRICT POLICY ON CHEATING THROUGH ANY SOURCE. ZERO MARKS WILL BE ASSIGNED IN ALL ASSIGNMENTS AND QUIZZES.
6. **Submission Deadline:** 06th November 2024.
7. **Submission Procedure:** Assignments should be placed on the teacher's table before the start of the class. Half marks will be deducted on providing the assignment during the class. No assignment will be accepted after the class in any case.
8. **Identification Details:** Ensure that your Roll Number, Section, Name, and Signature are written at the bottom of each page. Failure to do so will result in mark deductions, and no subsequent queries about this matter will be entertained.
9. **Solution Methodology:**
  - a. Understand the question and provide solution of each problem with complete steps.
  - b. All steps and calculations must be clearly mentioned.
  - c. If there are multiple solutions or approaches, use one of them to solve your problem but discuss the rest of the approaches to the problem as well
  - d. Draw neat and clean figures and tables where appropriate.
10. **Clarity and Neatness:** Your handwriting must be legible and neat. Any illegible or messy submissions may result in mark deductions.

## Section A: Robot Motion - Polynomial Trajectory

### Question # 1

- Given: A robot joint moves from  $\pi/3$  to  $7\pi/4$  rad over 4s, starting from rest and then stopping.
- Find: An appropriate quintic polynomial trajectory.
- Instructions: Create Quintic polynomial trajectory. Find  $q$ ,  $v$  and  $a$ . Find the inverse of  $6 \times 6$  matrix using MATLAB. (You are not allowed to convert it to a matrix of  $3 \times 3$  for this particular problem) Write the code in your assignment and put the screenshot of output on a separate page attached to this assignment. (Only the screenshot of output will be printed.)

### Question # 2

- Given: A robot joint moves from  $\pi/2$  to  $3\pi/2$  rad over 6s, starting from rest and then stopping.
- Find: An appropriate quintic polynomial trajectory.
- Instructions: Create Quintic polynomial trajectory. Find  $q$ ,  $v$  and  $a$ . You may convert it to a  $3 \times 3$  matrix. You are not allowed to use MATLAB for this problem. Write the inverse by hand. Inverse from calculator is not allowed either.

### Question # 3

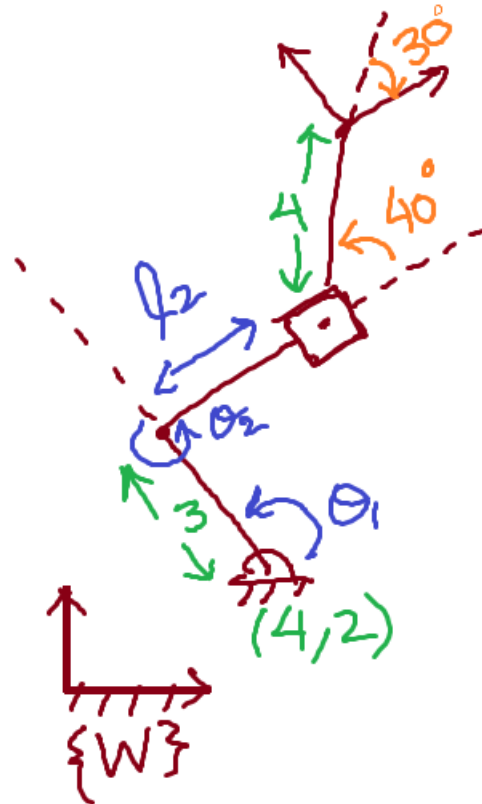
- Given: A robot joint moves from 0 to  $\pi/2$ . It starts this journey at 3s and ends at 8s. When it started this journey, it was already in motion. So at 3s its velocity was  $2\text{ nms}^{-1}$  and at 8s it acquired a constant velocity of  $5\text{ nms}^{-1}$ .
- Find: An appropriate quintic polynomial trajectory.
- Instructions: Create Quintic polynomial trajectory. Find  $q$ ,  $v$  and  $a$ . If you use MATLAB for calculating the inverse, then write code and attach screenshot of output as well. (only output can be printed).

## Section B: Inverse Kinematics

### Question # 1

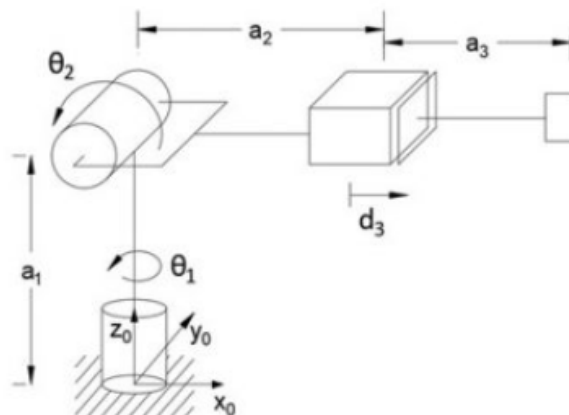
Given a planar robot's end-effector to be at Pose(4.7343, -3.8242, 160 degree) where the coordinates are (x, y, theta) relative to the fixed reference frame F.

Determine the required robot parameters from to achieve it. Robot is given in the figure.



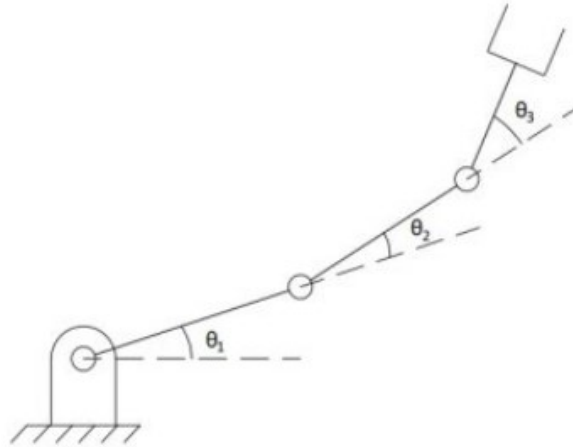
### Question # 2

An RRP robot's end-effector is at Pose(3, 4, 0 degree). Determine the required robot parameters  $\theta_1$ ,  $\theta_2$ , and  $d_3$  to achieve it.  $a_1=5$ ,  $a_2=5$ ,  $a_3=4$ . Robot is given in the figure. Use dh table for obtaining Transformation matrices.



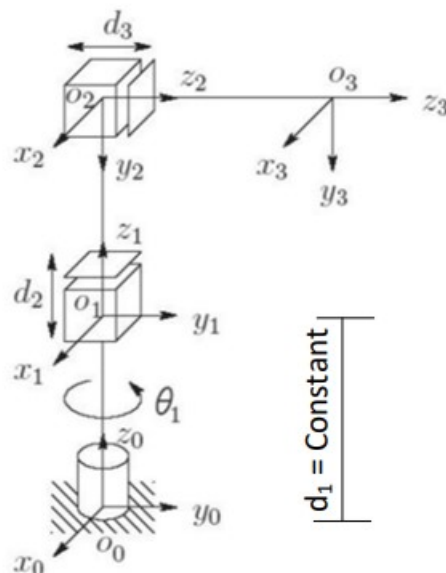
### Question # 3

Given the following 3 joint planar revolute arm, find the robot parameters when the desired position of end effector is at (5, 7, 45 degrees) which are (x, y, and theta)



### Question # 4

An RPP 3 link cylindrical manipulator is given. Perform inverse kinematics using algebraic approach on it by finding out the value of robot parameters  $\theta_1$ ,  $d_2$  and  $d_3$ . Write the answer in generalize form in terms of known parameters x, y, z and  $d_1$ . Hint: First perform FK to get the final T matrix.



### **Section 3: Controlling your hardware through mobile App using ESP8266**

**Question:** You are required to make wedding lights which can be controlled by your phone. Design a circuit with at least 5 LEDs and an ESP8266. Write program and use local host for a very simple mobile app with just three buttons. Make three different patterns of wedding light LEDs. By pressing each button, a different pattern should start running on your circuit. Attach code, console output and picture of your circuit here. Live demo of this circuit will also be checked later. (you are allowed to attach a printed copy of the solution of this task).

**--- The End ---**